# Analytic–Numerical Framework for Intraband Optical Properties of Lens-Shaped Quantum Dots

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## ABSTRACT

An analytic–numerical model is presented for the intraband optical properties of electrons confined in a lens-shaped (semi-ellipsoidal) quantum dot within the effective-mass and envelope-function approximations. Assuming hard-wall confinement, the single-particle wave functions and energies are obtained using adiabatic separation of the fast (vertical) and slow (in-plane) motions. The linear and third-order nonlinear intraband absorption coefficients and refractive-index changes are calculated using the density-matrix formalism. The influence of the external magnetic field is analyzed through its effect on the energy spectrum and optical response. Numerical results show that the dot's geometric parameters strongly modulate the linear and nonlinear response, while the effect of the magnetic field remains weak for the considered dot sizes. For representative GaAs-like parameters, single-particle energies are much more sensitive to the dot height than to the lateral semi-axis. States with higher longitudinal quantum numbers lie well above the lowest branch. Nonlinear effects lead to a reduction and reshaping of the absorption resonances. Magnetic-field-induced modifications become significant only for larger quantum dots, where the cyclotron energy is comparable to the confinement energy.

**Keywords:** Lens-shaped quantum dot,Intraband absorption, Nonlinear optical properties, Hard-wall confinement

## 1. INTRODUCTION

Semiconductor quantum dots (QDs) are nanoscale systems in which charge carriers are confined in all three spatial dimensions, resulting in discrete, atom-like energy spectra and highly tunable optical properties. Owing to these features, QDs have found widespread applications in optoelectronic devices such as light-emitting diodes and lasers, in photovoltaics through bandgap engineering and multiple exciton generation, and in quantum information technologies as sources of single and entangled photons. In addition, their high photostability and tunable emission make them attractive for biomedical imaging and sensing. More recently, their strong light-matter interaction and pronounced nonlinear response have positioned quantum dots as promising platforms for infrared and terahertz photonics, including photodetectors and optical modulators. These diverse applications stem from the ability to engineer their electronic structure via size, shape, and external fields.[1,2]

A central factor governing the electronic structure and optical response of quantum dots is their geometrical shape. Different confinement profiles lead to distinct energy spectra, selection rules, and spatial distributions of wavefunctions. Among various geometries, ellipsoidal quantum dots have attracted considerable attention due to their anisotropic confinement, which allows independent control of longitudinal and lateral motion. Optical properties of such systems, including direct interband absorption, have been extensively studied for both strongly oblate and strongly prolate geometries.[3,4] Extensions of these models have incorporated many-body effects such as biexcitons and trions,[5] as well as ensemble behavior.[6] Furthermore, the influence of confinement geometry on collective excitations has been explored through the realization of Kohn's theorem in ellipsoidal quantum dots.[7,8]

More generally, optical processes in quantum dots—including absorption, Raman scattering, and optical gain—have been investigated for a wide range of confinement geometries.[9–11] These studies highlight that

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even subtle changes in geometry can lead to significant modifications of the optical response, emphasizing the importance of accurate modeling of realistic quantum dot shapes.

Semiellipsoidal (lens-shaped) quantum dots are of particular experimental relevance, as they closely resemble structures formed via self-assembled growth techniques. Their strong anisotropy leads to a natural separation between longitudinal and planar motion, while simultaneously inducing nontrivial coupling between these degrees of freedom. This coupling manifests as an effective in-plane confinement potential that modifies both the energy spectrum and the spatial structure of the wavefunctions. Although absorption processes have been studied in other geometries, such as conical and cylindrical quantum dots,[12,13] the combined effect of semiellipsoidal geometry, magnetic field, and nonlinear optical response remains relatively less explored, particularly for intraband transitions.

In this work, we investigate the intraband optical response of semi-ellipsoidal quantum dots in the presence of a magnetic field. Using an adiabatic approach, we derive an effective description that captures both geometric confinement and magnetic effects, leading to a Fock–Darwin-type spectrum modified by the dot geometry. The optical properties are analyzed within the density matrix formalism, allowing us to obtain both linear and nonlinear susceptibilities, as well as the corresponding absorption and refractive index changes. Our results demonstrate that the optical response can be efficiently tuned via geometry, magnetic field, and excitation intensity, highlighting the potential of such systems for controllable infrared and terahertz photonic applications.

# 2. THEORY

## 2.1 Semi-ellipsoidal Quantum Dot

We consider a semiconductor quantum dot with a lens-shaped geometry, modeled as a semi-ellipsoid confined to the half-space $z \geq 0$. The domain of the particle is defined as

$$\mathcal{D} = \left\{ (x, y, z) : \frac{x^2 + y^2}{a^2} + \frac{z^2}{c^2} \leq 1, \quad z \geq 0 \right\}, \tag{1}$$

where $a$ and $c$ denote the lateral and vertical semi-axes, respectively. Throughout this work, we focus on strongly flattened quantum dots satisfying

$$\frac{c}{a} \ll 1. \tag{2}$$

This strong anisotropy leads to a separation between longitudinal and lateral energy scales.

The confinement of the charge carrier is modeled by an infinite potential barrier at the boundary of the domain (1),

$$V_{\text{conf}}(x, y, z) = \begin{cases} 0, & (x, y, z) \in \mathcal{D}, \\ \infty, & \text{otherwise}, \end{cases} \tag{3}$$

which enforces the boundary condition

$$\Psi\big|_{\partial\mathcal{D}} = 0. \tag{4}$$

This hard-wall confinement causes complete localization of the particle within the quantum dot.

An external magnetic field is applied along the growth direction,

$$\mathbf{B} = B\hat{\mathbf{z}}, \tag{5}$$

which preserves rotational symmetry in the lateral plane. To exploit this symmetry, we adopt the symmetric gauge for the vector potential,

$$\mathbf{A} = \frac{B}{2}(-y, x, 0). \tag{6}$$

Within the effective-mass approximation, the single-particle Hamiltonian takes the form

$$H_0 = \frac{1}{2m^*} \left(\mathbf{p} + q\mathbf{A}\right)^2 + V_{\text{conf}}(x, y, z), \tag{7}$$

where $m^*$ is the effective mass and $q$ is the carrier charge.

### 2.2 Adiabatic Approximation and Planar Spectrum

The strong geometric anisotropy expressed by Eq. (2) implies that the motion along the growth direction is much faster than the in-plane motion. This allows us to employ an adiabatic approximation, in which the longitudinal and planar degrees of freedom are separated.

At each radial position $r = \sqrt{x^2 + y^2}$, the confinement along the $z$-direction is determined by the local height of the semi-ellipsoid,

$$z_{\max}(r) = c\sqrt{1 - \frac{r^2}{a^2}}. \tag{8}$$

The motion along $z$ is therefore equivalent to that of a particle in an infinite potential well of width $z_{\max}(r)$, leading to eigenfunctions

$$\phi_\nu(z; r) = \sqrt{\frac{2}{z_{\max}(r)}} \sin\left(\frac{\nu\pi z}{z_{\max}(r)}\right), \tag{9}$$

and corresponding energies

$$\varepsilon_\nu(r) = \nu^2 \varepsilon_z \frac{1}{1 - r^2/a^2}, \qquad \varepsilon_z = \frac{\hbar^2\pi^2}{2m^* c^2}. \tag{10}$$

The key point is that the longitudinal energy $\varepsilon_\nu(r)$ depends explicitly on the radial coordinate. This induces an effective potential for the in-plane motion. Expanding Eq. (10) for $r \ll a$, one obtains

$$\varepsilon_\nu(r) \approx \nu^2\varepsilon_z + \nu^2\varepsilon_z \frac{r^2}{a^2}, \tag{11}$$

which shows that the geometric confinement along $z$ generates an additional parabolic potential in the plane.

Combining this contribution with the magnetic field terms in Eq. (7), the effective planar Hamiltonian for each subband $\nu$ becomes

$$H^{(\nu)} = \frac{p_x^2 + p_y^2}{2m^*} + \frac{\omega_c}{2} L_z + \left(\frac{m^*\omega_c^2}{8} + \frac{\nu^2\varepsilon_z}{a^2}\right) r^2, \tag{12}$$

where $L_z = xp_y - yp_x$ and $\omega_c = qB/m^*$ is the cyclotron frequency.

The magnetic field affects the planar motion in two distinct ways: it introduces a coupling to the angular momentum through the term $\frac{\omega_c}{2} L_z$, and it renormalizes the effective confinement via the quadratic term proportional to $\omega_c^2$. As a result, the in-plane dynamics corresponds to a two-dimensional harmonic oscillator in a magnetic field, i.e., the Fock–Darwin problem.

The effective confinement frequency is therefore given by

$$\omega_\nu = \sqrt{\frac{2\nu^2\varepsilon_z}{m^* a^2} + \frac{\omega_c^2}{4}}, \tag{13}$$

and the energy spectrum takes the form

$$E_{\nu,n,m} = \nu^2\varepsilon_z + \hbar\omega_\nu(2n + |m| + 1) - \frac{\hbar\omega_c}{2} m. \tag{14}$$

The corresponding planar eigenfunctions are

$$\psi_{n,m}(r, \varphi) = \mathcal{N} r^{|m|} e^{-r^2/(2\ell_\nu^2)} L_n^{|m|}\left(\frac{r^2}{\ell_\nu^2}\right) e^{im\varphi}, \tag{15}$$

where $\mathcal{N}$ is a normalization coefficient, $L_n^{|m|}$ are associated Laguerre polynomials, and

$$\ell_\nu = \sqrt{\frac{\hbar}{m^*\omega_\nu}}. \tag{16}$$

The total wave function is then written as

$$\Psi_{\nu,n,m}(z, r, \varphi) = \phi_\nu(z; r)\, \psi_{n,m}(r, \varphi), \tag{17}$$

which forms the basis for the calculation of optical transition matrix elements.

### 2.3 Light–Matter Interaction and Quantum Dynamics

We now consider the interaction of the quantum dot with an external electromagnetic field polarized along the growth direction. The electric field is taken in the form

$$E_z(t) = E_\omega e^{-i\omega t} + E_\omega^* e^{i\omega t}, \tag{18}$$

where $E_\omega$ is the complex amplitude and $\omega$ is the optical frequency.

Within the dipole approximation, the interaction Hamiltonian is given by

$$H_{\text{int}}(t) = -qzE_z(t), \tag{19}$$

which couples quantum states through the matrix elements of the position operator along the confinement direction.

The time evolution of the system is described using the density matrix formalism. The density operator $\rho$ satisfies the Liouville equation

$$\dot{\rho} = -\frac{i}{\hbar}[H_0 + H_{\text{int}}(t), \rho] - \Gamma(\rho), \tag{20}$$

where $\Gamma(\rho)$ accounts phenomenologically for relaxation and dephasing processes. In the present model, we assume a single relaxation rate,

$$\Gamma(\rho)_{ij} = \Gamma\,(\rho_{ij} - \rho_{ij}^{(0)}), \tag{21}$$

which drives the system toward equilibrium populations.

Projecting Eq. (20) onto the basis of eigenstates $|i\rangle = |\nu, n, m\rangle$, one obtains the equations of motion for the density matrix elements,

$$\dot{\rho}_{ij} = -(i\omega_{ij} + \Gamma)\rho_{ij} + \frac{i}{\hbar} d_{ij} q E_z(t)\,(\rho_{jj} - \rho_{ii}), \tag{22}$$

where the transition frequency is defined as

$$\hbar\omega_{ij} = E_j - E_i, \tag{23}$$

and the dipole matrix elements are

$$d_{ij} = \langle i|z|j\rangle. \tag{24}$$

The dipole matrix elements $d_{ij} = \langle i|z|j\rangle$ impose selection rules that follow from the symmetry of the system. Since the interaction operator $z$ acts only along the growth direction, it does not affect the in-plane angular variables. As a result, the angular momentum quantum number is conserved,

$$\Delta m = 0. \tag{25}$$

In contrast, transitions along the $z$-direction are governed by the structure of the confined states $\phi_\nu(z)$, allowing changes in the longitudinal quantum number $\nu$. Similarly, the radial quantum number $n$ is not strictly restricted by symmetry.

Although in principle, transitions with arbitrary $\Delta\nu$ and $\Delta n$ are allowed, the dominant contribution arises from transitions between neighboring longitudinal subbands with minimal change in the planar motion. Therefore, the leading optical transitions satisfy

$$\Delta\nu = \pm 1, \qquad \Delta n = 0, \qquad \Delta m = 0, \tag{26}$$

which correspond to intraband excitations primarily along the growth direction.

### 2.4 Two-Level Approximation and Optical Response

In order to describe the optical response, we restrict the system to an effective two-level model consisting of the ground state $|0\rangle$ and a selected excited state $|1\rangle$. This reduction is based on the assumption that the optical transitions of interest originate from the ground state, while transitions between excited states are neglected.

The density matrix is expanded perturbatively in powers of the electric field,

$$\rho = \rho^{(0)} + \rho^{(1)} + \rho^{(3)} + \cdots , \tag{27}$$

where the zeroth-order term corresponds to equilibrium,

$$\rho_{00}^{(0)} = 1, \qquad \rho_{11}^{(0)} = 0. \tag{28}$$

Solving Eq. (22) to first order in the field yields the linear coherences

$$\rho_{01}^{(1)}(\omega) = \frac{d_{01}E_\omega}{\hbar\omega_{10} + \hbar\omega + i\Gamma}. \tag{29}$$

Higher-order terms give rise to nonlinear effects. In particular, the third-order coherence scales as

$$\rho_{01}^{(3)} \propto E_\omega |E_\omega|^2, \tag{30}$$

which is responsible for intensity-dependent optical properties.

### 2.5 Macroscopic Polarization and Susceptibilities

The optical response of the quantum dot ensemble is described by the macroscopic polarization,

$$P_z = \sigma \sum_{ij} \rho_{ji} d_{ij}, \tag{31}$$

where $\sigma$ is the quantum dot density.

From the linear contribution, one obtains the first-order susceptibility,

$$\chi^{(1)}(\omega) = \frac{\sigma |d_{01}|^2}{\varepsilon_0 \hbar} \frac{1}{\omega_{10} - \omega - i\Gamma}, \tag{32}$$

which describes resonant absorption and dispersion.

The third-order contribution leads to a nonlinear susceptibility of the form[14]

$$\mathrm{Re}\,\chi^{(3)}(\omega) = -\frac{2\sigma}{\varepsilon_0\hbar^3} \frac{|d_{01}|^4(\omega_{10}-\omega)}{\left((\omega_{10}-\omega)^2+\Gamma^2\right)^2} \left(1 - \frac{|d_{11}-d_{00}|^2}{4|d_{01}|^2} \frac{\omega_{10}(\omega_{10}-\omega)^2 - \Gamma^2(3\omega_{10}-2\omega)}{(\omega_{10}^2+\Gamma^2)(\omega_{10}-\omega)}\right) \tag{33}$$

$$\mathrm{Im}\,\chi^{(3)}(\omega) = -\frac{2\sigma}{\varepsilon_0\hbar^3} \frac{|d_{01}|^4\Gamma}{\left((\omega_{10}-\omega)^2+\Gamma^2\right)^2} \left(1 - \frac{|d_{11}-d_{00}|^2}{4|d_{10}|^2} \frac{(3\omega_{10}-\omega)(\omega_{10}-\omega) - \Gamma^2}{\omega_{10}^2+\Gamma^2}\right). \tag{34}$$

which captures the nonlinear optical response, including saturation and intensity-dependent refractive effects.

### 2.6 Optical Coefficients

The intensity of the electromagnetic field is related to the field amplitude through

$$I = \frac{1}{2} n_r c \varepsilon_0 |E_\omega|^2, \tag{35}$$

which allows the total susceptibility to be written as

$$\chi(\omega) = \chi^{(1)}(\omega) + \chi^{(3)}(\omega) \frac{2I}{n_r c \varepsilon_0}. \tag{36}$$

The absorption coefficient is given by

$$\alpha(\omega) = \frac{\omega}{n_r c} \operatorname{Im} \chi(\omega), \tag{37}$$

while the change in refractive index is

$$\Delta n(\omega) = \frac{1}{2n_r} \operatorname{Re} \chi(\omega). \tag{38}$$

## 3. RESULTS

### 3.1 Spectrum and Wavefunctions

The structure of the eigenstates is illustrated in Fig. 1, where representative probability densities are shown for several combinations of quantum numbers $(\nu, n, m)$. The wavefunctions clearly reflect the separation between longitudinal and planar motion introduced in Eq. (17).

The longitudinal quantum number $\nu$ controls the number of nodes along the growth direction. Increasing $\nu$ introduces additional nodes in the vertical profile and raises the confinement energy. In contrast, the radial quantum number $n$ governs the number of nodes in the in-plane radial direction, while the angular momentum $m$ determines the azimuthal structure.

For $m = 0$, the states remain rotationally symmetric, whereas finite $m$ leads to ring-like structures with angular modulation. These features are consistent with the Fock–Darwin spectrum in Eq. (14). The magnetic field lifts degeneracies and modifies the spatial distribution, particularly for states with nonzero angular momentum.

Importantly, the wavefunction structure directly determines the dipole matrix elements. Since the interaction operator $z$ acts only along the growth direction, the dominant transitions involve changes in $\nu$ while preserving the in-plane quantum numbers, in agreement with Eq. (26).

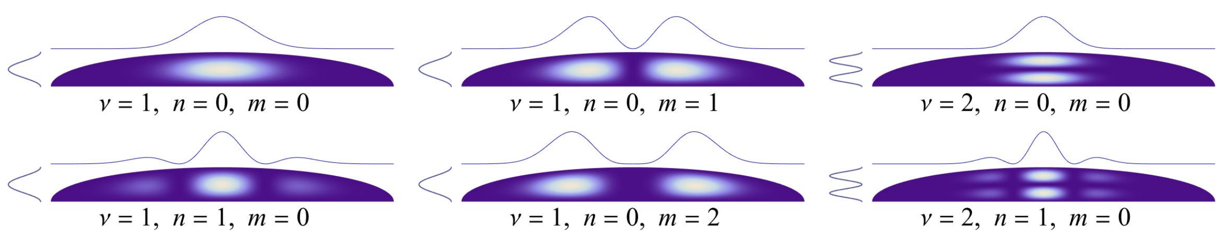


Figure 1. Representative probability densities for selected quantum states $(\nu, n, m)$ in a semi-ellipsoidal quantum dot. The plots illustrate the separation between longitudinal and planar motion.

### 3.2 Dipole Matrix Elements

The calculated dipole matrix elements for different geometrical parameters are summarized in Table 1. The results confirm that the strongest transitions correspond to $\Delta\nu = 1$ with minimal changes in the planar quantum numbers, in agreement with the selection rules in Eq. (26).

Variations in the geometrical parameters $a$ and $c$ affect the dipole moments in qualitatively different ways. The dependence on the lateral size $a$ is generally weak: only the transition $|1, 0, 0\rangle \rightarrow |1, 1, 0\rangle$ shows noticeable sensitivity, while higher transitions involving changes in $\nu$ remain nearly unchanged. This can be attributed to the fact that these transitions are dominated by the longitudinal overlap of the wavefunctions, with only a minor contribution from the planar part.

In contrast, the dipole moments exhibit a strong dependence on the vertical size $c$. Since the dipole operator $z$ acts along the growth direction, the matrix elements are primarily determined by the structure of the longitudinal wavefunctions $\phi_\nu(z)$, which scale directly with the confinement length. As a result, even small variations in $c$ lead to significant changes in the transition strengths for all considered transitions.

In contrast, the effect of the magnetic field on the dipole moments is found to be negligible in the considered parameter range. This behavior can be understood from the structure of the matrix elements $d_{ij} = \langle i|z|j\rangle$, which depend primarily on the longitudinal wavefunctions $\phi_\nu(z)$. Since the magnetic field enters only through the planar Hamiltonian, its influence is limited to the in-plane wavefunctions $\psi_{n,m}(r,\varphi)$, while leaving the dominant longitudinal overlap essentially unchanged.

Furthermore, for the quantum dot sizes considered here, the magnetic confinement energy $\hbar\omega_c$ remains small compared to the effective confinement scale $\hbar\omega_\nu$. As a result, the magnetic field induces only weak modifications of the planar wavefunctions, leading to negligible corrections to the dipole matrix elements. Consequently, the dipole moments can be considered effectively independent of the magnetic field in this regime.

Table 1. Numerical values of the $z$-dipole moments for selected transitions.

| $a$, nm | $c$, nm | $\lvert 1,0,0\rangle \to \lvert 1,1,0\rangle$ | $\lvert 1,0,0\rangle \to \lvert 2,0,0\rangle$ | $\lvert 1,0,0\rangle \to \lvert 2,1,0\rangle$ | $\lvert 1,0,0\rangle \to \lvert 2,2,0\rangle$ |
|---|---|---|---|---|---|
| 20 | 3.8 | 0.06 | -0.63 | -0.22 | -0.08 |
| 10 | 4.0 | 0.15 | -0.65 | -0.24 | -0.09 |
| 15 | 4.0 | 0.09 | -0.66 | -0.23 | -0.08 |
| 20 | 4.0 | 0.07 | -0.66 | -0.23 | -0.08 |
| 30 | 4.0 | 0.04 | -0.67 | -0.23 | -0.08 |
| 20 | 5.0 | 0.11 | -0.83 | -0.29 | -0.10 |
| 20 | 6.0 | 0.16 | -0.99 | -0.35 | -0.12 |

### 3.3 Absorption and Refractive Index

The absorption coefficient, calculated from Eq. (37), is shown in Fig. 2, while the corresponding refractive index change, obtained from Eq. (38), is presented in Fig. 3. Both quantities exhibit clear signatures of the underlying quantized energy spectrum.

All calculations were performed using GaAs material parameters. The phenomenological relaxation parameter was taken as $\Gamma = 5$ meV, and the incident optical intensity was fixed at $I = 5 \times 10^{10}$ W/m$^2$.

The absorption spectra exhibit pronounced resonant peaks corresponding to transitions between quantized energy levels. The linear contribution dominates at low intensities and produces symmetric resonances centered at the transition energies. The nonlinear contribution introduces a negative correction near resonance, leading to a reduction of the peak amplitude and a modification of the spectral shape. This behavior reflects saturation effects characteristic of two-level systems and becomes noticeable at the chosen intensity.

The refractive index change displays the corresponding dispersive behavior, with a characteristic sign change across each resonance. The nonlinear contribution modifies both the amplitude and the slope of the dispersion curve, leading to intensity-dependent refractive properties. These features directly follow from the interplay between the real and imaginary parts of the susceptibility.

Geometrical parameters influence the optical response in distinct ways. Variations in the lateral size $a$ primarily affect transitions involving excited planar states, modifying the higher-energy features of both absorption and dispersion spectra. In contrast, the position of the main resonance is highly sensitive to the vertical size $c$, since the dominant transition is governed by the longitudinal confinement. As a result, small changes in $c$ lead to significant shifts in both the absorption peak and the corresponding dispersive feature.

In contrast, the effect of the magnetic field on both absorption and refractive index is weak for the quantum dot sizes considered in this work. This is due to the relatively small cyclotron energy compared to the confinement energy, such that the magnetic field induces only minor modifications of the effective in-plane potential. Noticeable magnetic-field-induced changes are expected only for significantly larger quantum dots, with characteristic sizes on the order of $\sim 100$ nm.

Overall, the results demonstrate that the optical response of semi-ellipsoidal quantum dots is primarily governed by geometrical confinement and excitation intensity, while magnetic field effects become relevant only in larger structures. This provides a flexible platform for tuning both absorptive and dispersive properties in the infrared regime.

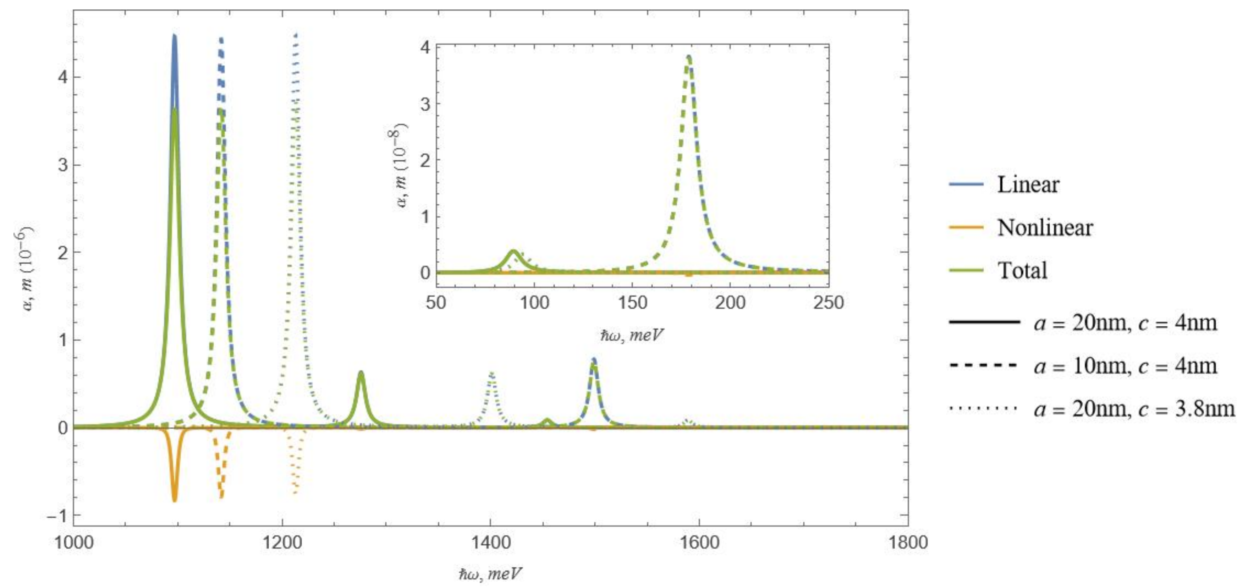


Figure 2. Absorption coefficient as a function of photon energy for different geometrical parameters and magnetic field strengths. Linear, nonlinear, and total contributions are shown, illustrating the role of saturation and intensity-dependent effects.

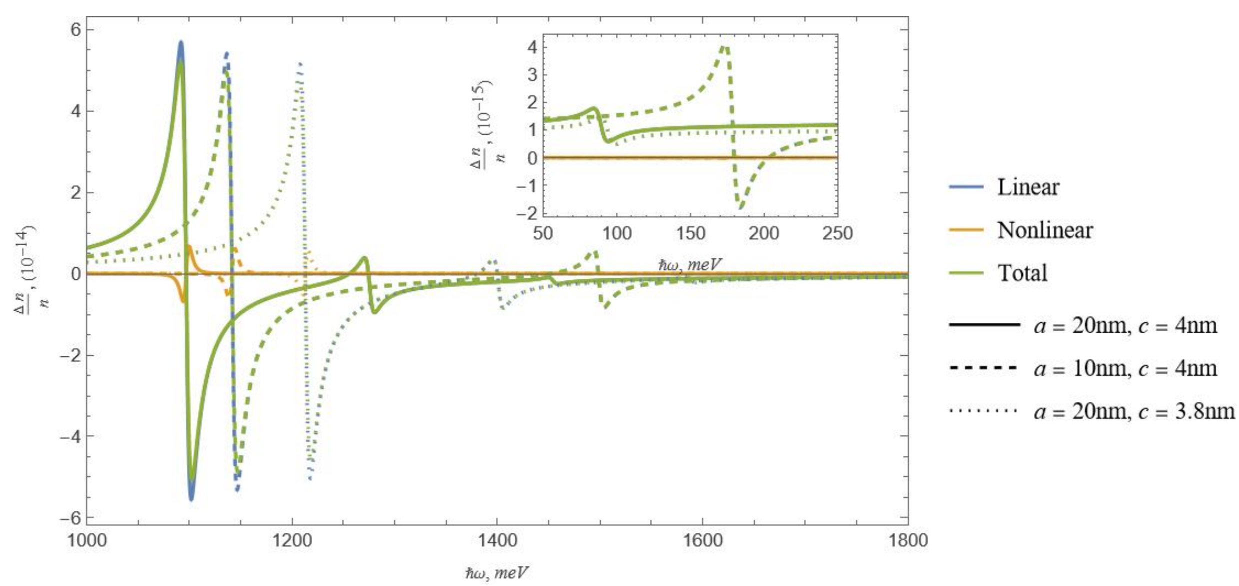


Figure 3. Refractive index change as a function of photon energy. The dispersive behavior follows the absorption spectrum and highlights the impact of nonlinear optical contributions and magnetic field effects.

## 4. CONCLUSION

In this work, we have investigated the intraband optical properties of semi-ellipsoidal quantum dots within an effective-mass framework, taking into account both geometrical confinement and external magnetic fields. By employing an adiabatic approach, the three-dimensional problem was reduced to an effective planar system, leading to a modified Fock–Darwin spectrum that captures the interplay between confinement and magnetic effects.

The calculated energy spectrum and wavefunctions reveal a clear separation between longitudinal and planar motion, which directly determines the optical selection rules and transition strengths. In particular, the dipole matrix elements are governed primarily by the longitudinal confinement, resulting in a strong sensitivity to the vertical size of the quantum dot, while remaining largely insensitive to the magnetic field in the considered parameter range.

The analysis of the optical response, performed within the density matrix formalism, shows that both absorption and refractive index are strongly influenced by geometrical parameters and excitation intensity. The dominant absorption peak, associated with transitions between longitudinal states, exhibits a high sensitivity to the dot height, whereas variations in the lateral size primarily affect transitions involving excited planar modes. Nonlinear effects lead to a reduction of absorption peaks and a modification of the dispersive response, reflecting saturation behavior characteristic of two-level systems.

The influence of the magnetic field was found to be weak for quantum dots of typical sizes considered in this work. This is explained by the smallness of the cyclotron energy compared to the confinement energy, such that magnetic-field-induced modifications of the optical response become noticeable only for significantly larger structures.

Overall, the results demonstrate that the optical properties of semi-ellipsoidal quantum dots are predominantly governed by geometrical confinement and can be efficiently tuned through structural design and excitation intensity. This makes such systems promising candidates for controllable infrared and terahertz optoelectronic applications, including photodetectors, modulators, and nonlinear optical devices.

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